## EXPERIMENT 9: AI AND DATA ANALYTICS USING ZOHO ZIA IN ZOHO CREATOR

### AIM

To implement AI and data analytics using Zoho Zia in Zoho Creator to analyze student attendance and academic performance data, identify trends, and generate insights to support academic monitoring and decision-making.

### PROCEDURE

1. Log in to Zoho Creator and select the option to create an application using Zia, if available in the account.
2. Describe the requirement as a Student Attendance and Academic Performance Tracker application.
3. Review the use cases and suggested data model, including Student Details, Attendance Records, and Academic Performance.
4. Create the application and review the generated forms, reports, and dashboard.
5. Add sample student details, attendance records, and academic marks so that the dashboard has data to analyze.
6. Open the application editor and use the available Zia Build Agent or AI-assisted building feature to modify the application, if supported.
7. Open the Attendance Report and Academic Performance Report to review student attendance percentages and examination marks.
8. Study the charts and reports to identify attendance patterns, subject-wise performance, and students who may need academic support.
9. Verify the generated insights against the actual records to ensure accuracy.

### SAMPLE PROMPTS GIVEN TO ZIA

- "Create an application to manage student details, daily attendance, assignment marks, test marks, and final examination results."
- "Add a dashboard card showing the total number of students and the average attendance percentage."
- "Create a report that identifies students whose attendance is below 75%."
- "Add a chart showing the distribution of students by academic performance."
- "Create a report listing students whose final marks are below 40."
- "Add a dashboard displaying subject-wise average marks and attendance trends."

## PROCESS DIAGRAM

**Figure 9.1: AI and Data Analytics Workflow**

Student Details, Attendance and Marks
↓
Zoho Creator Forms and Database
↓
Zia AI-Assisted Application Building (if available)
↓
Reports and Dashboard
↓
Data Analysis and Trend Identification
↓
Academic Monitoring and Decision-Making

## DASHBOARD METRICS GENERATED FROM SAMPLE DATA

The following values are illustrative expected outputs for the five sample students used in the project report.

| Metric | Value |
|---|---|
| Total students | 5 |
| Students in AIML | 5 |
| Average final mark | 73.6 |
| Students with attendance below 75% | 2 |
| Students with final marks below 40 | 0 |

Note: The metrics depend on the actual records entered into Zoho Creator. The average final mark is calculated from the sample final marks of 82, 90, 48, 86, and 62.

## ACADEMIC PERFORMANCE ANALYSIS (SAMPLE)

| Student | Attendance | Final Mark | Observation |
|---|---|---|---|
| Arun | 90% | 82 | Good performance |
| Priya | 75% | 90 | High academic performance |
| Karthik | 60% | 48 | Attendance improvement needed |
| Divya | 85% | 86 | Good performance |

| | | | |
|---|---|---|---|
| Rahul | 70% | 62 | Attendance improvement needed |

**TRENDS IDENTIFIED**

1. Students with attendance below 75% can be identified for attendance monitoring.
2. Students with lower examination marks may benefit from additional academic support.
3. Comparing attendance and marks helps faculty examine possible relationships between participation and performance.
4. Subject-wise charts can identify subjects in which students may require additional practice.
5. Dashboard reports help faculty monitor student progress and plan suitable interventions.

**RESULT**

AI-assisted application building and data analytics were explored for the Student Attendance and Academic Performance Tracker using Zoho Creator. The application data can be visualized through reports and dashboards to identify attendance patterns and academic performance trends.

**CONCLUSION**

Zoho Creator provides tools for developing applications, analyzing records, and presenting insights through reports and dashboards. AI-assisted features, where available, can simplify application creation and modification. By analyzing student attendance and academic marks, the application supports academic monitoring and helps faculty identify students who may need additional support.

Actual AI-generated predictions require a supported AI feature or an integrated machine learning service. Basic counts, averages, threshold